Real Time systems using free RTOS

(seat heater control system project)

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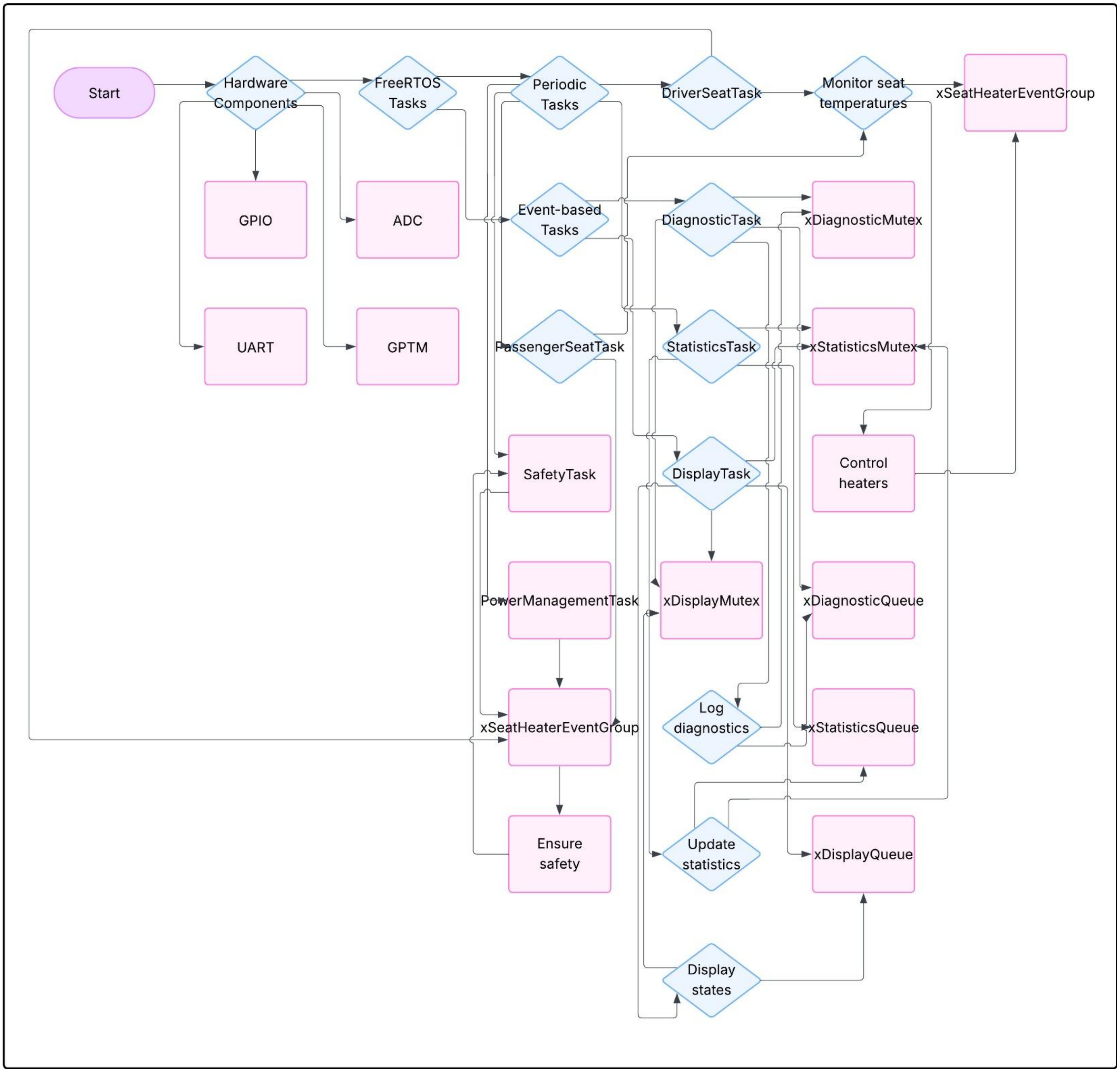
Table of contents:

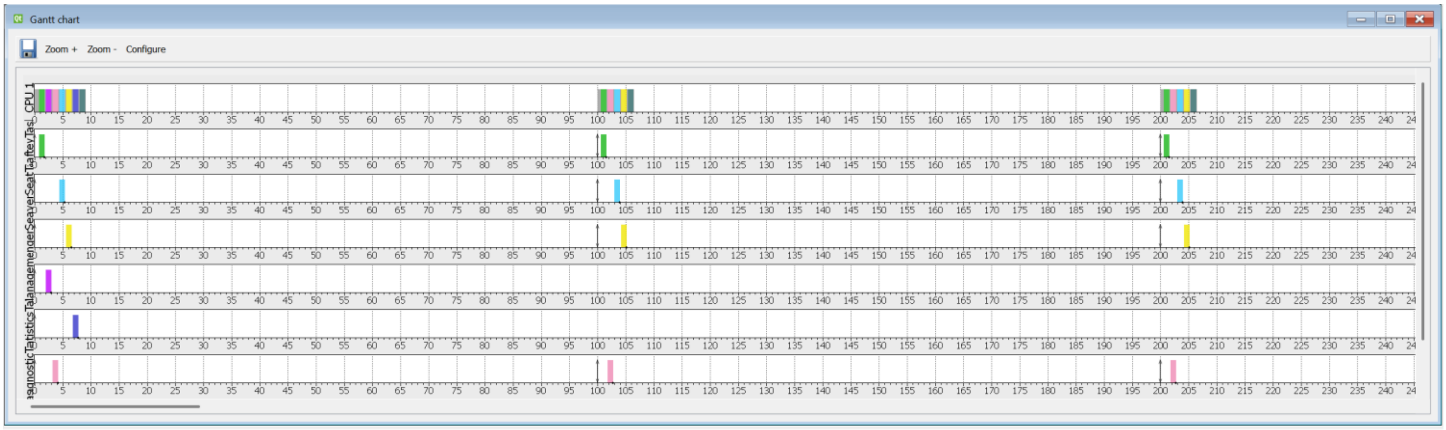
1- System design diagram:



2- List of shared resources:

RESOURCE	TASKS SHRAING IT	ACCESS METHOD
xDisplayQueue	DriverSeatTask, PassengerSeatTask, DisplayTask	Queue (xQueueSend / xQueueReceive)
xDiagnosticQueue	DriverSeatTask, PassengerSeatTask, DiagnosticTask	Queue (xQueueSend / xQueueReceive)
xStatisticsQueue	DriverSeatTask, PassengerSeatTask, StatisticsTask	Queue (xQueueSend / xQueueReceive)
xDisplayMutex	DisplayTask	Mutex (xSemaphoreTake / xSemaphoreGive)
xDiagnosticMutex	DiagnosticTask	Mutex (xSemaphoreTake / xSemaphoreGive)
xStatisticsMutex	StatisticsTask	Mutex (xSemaphoreTake / xSemaphoreGive)
xSeatHeaterEventGroup	GPIOF_Handler, DiagnosticTask	Event group (xEventGroupSetBitsFromISR/ xEventGroupWaitBits)
driverSeatState	DriverSeatTask, DisplayTask, DiagnosticTask, SafetyTask	————
passengerSeatState	PassengerSeatTask, DisplayTask, DiagnosticTask, SafetyTask	————
diagnosticLog	DiagnosticTask	Mutex (xDiagnosticMutex)

3- SIMSO SIMULATION RESULTS:



Results

General | Logs | Tasks | Scheduler | Processors

General | SafteyTask | DriverSeatTask | PassengerSeatTask | Powe

Computation time:

Task	min	avg	max	std dev	occupancy
SafteyTask	1.000	1.000	1.000	0.000	0.010
DriverSeatTask	1.000	1.000	1.000	0.000	0.010
PassengerSeatTask	1.000	1.000	1.000	0.000	0.010
PowerManagementTask	1.000	1.000	1.000	0.000	0.001
StatisticsTask	1.000	1.000	1.000	0.000	0.001

Preemptions:

Migrations:

Task migrations:

Response time:

Results

General | Logs | Tasks | Scheduler | Processors

Observation Window:

from 0.00 to 2000.00 ms

Configure...

	Total load	Payload	System load
CPU 1	0.0494	0.0380	0.0114
Average	0.0494	0.0380	0.0114

Results

General | Logs | Tasks | Scheduler | Processors

CPU 1

Cxt Save count: 76

Cxt Load count: 76

Cxt Save overhead: 3.8000ms (3800000 cycles)

Cxt Load overhead: 3.8000ms (3800000 cycles)

Results

General | Logs | Tasks | Scheduler | Processors

Scheduling events:

schedule count: 152

on_activate count: 77

on_terminate count: 76

Scheduling overhead:

schedule overhead: 7.6000ms (7600000 cycles)

on_activate overhead: 3.8000ms (3800000 cycles)

on_terminate overhead: 3.8000ms (3800000 cycles)

Sum: 15.2000ms (15200000 cycles)

Timers

CPU 1: 0

4- UART output screenshots:

```
=== Seat Heater ===

Driver:
Level: OFF
Temp: 12.4 C
Heat: OFF
Status: OK

Passenger:
Level: OFF
Temp: 13.6 C
Heat: OFF
Status: OK
```

Terminal ×

COM6 ×

```
=== Seat Heater ===

Driver:
Level: LOW
Temp: 18.9 C
Heat: HIGH
Status: OK

Passenger:
Level: OFF
Temp: 20.1 C
Heat: OFF
Status: OK


S:160 P:68 D:186
=== Seat Heater ===

D: 18.9 MED OK
P: 20.1 OFF OK

=== Seat Heater ===

Driver:
Level: MED
Temp: 18.9 C
Heat: HIGH
Status: OK

Passenger:
Level: OFF
Temp: 20.1 C
Heat: OFF
Status: OK


S:160 P:68 D:186
=== Seat Heater ===

Driver:
Level: HIGH
Temp: 18.9 C
Heat: HIGH
Status: OK

Passenger:
Level: OFF
Temp: 20.1 C
```

Terminal ×

COM6 ×

```
S:160 P:68 D:186
=== Seat Heater ===

Driver:
Level: OFF
Temp: 18.9 C
Heat: OFF
Status: OK

Passenger:
Level: LOW
Temp: 20.1 C
Heat: MED
Status: OK


S:160 P:68 D:186
=== Seat Heater ===

D: 18.9 OFF OK
P: 20.1 MED OK

=== Seat Heater ===

Driver:
Level: OFF
Temp: 18.9 C
Heat: OFF
Status: OK

Passenger:
Level: MED
Temp: 20.1 C
Heat: HIGH
Status: OK


S:160 P:68 D:186
=== Seat Heater ===

Driver:
Level: OFF
Temp: 18.9 C
Heat: OFF
Status: OK

Passenger:
```